

STATEMENT OF SPECIFICATIONS FOR THE PROCUREMENT OF A 350 TON Press Brake

**THE PURCHASE DESCRIPTION SET FORTH BELOW REFLECTS THOSE
CHARACTERISTICS WHICH ARE ESSENTIAL TO THE MINIMUM NEEDS OF THE
GOVERNMENT**

1. SPECIFICATIONS:

1.1. SCOPE: This specification covers the requirements for the procurement, installation and training of one (1) each 350 Ton Press Brake.

2. EQUIPMENT DESCRIPTION: The equipment shall consist of one 350 Ton Press Brake that meets the requirements below.

2.1 SALIENT CHARACTERISTICS: The equipment will have the following salient characteristics:

2.2. Press Capacity: 350 full tonnage.

2.3. Dimensions:

2.3.1. Machine length: 160” max.

2.3.2. Machine width: 75” max

2.3.3. Machine height: 150” max

2.3.4. Bed Length/Ram Length: 144” max.

2.3.5. Stroke length: 10” minimum.

2.4. Weight: 55000 lbs. maximum.

2.5. Frame: The frame is of open throat design with mounting holes in feet. Drilled and tapped holes in feet allow for leveling of the machine.

2.6. Bed: The machined flat surface is slotted full length to accommodate standard die holders, etc.

2.7. Ram: The ram is constructed of solid steel slotted to accept standard dies. The clamping bar is sectional for positive clamping. The upper is machined up to 4” to allow a flat surface for mounting brackets and gauges.

2.8. Ram Cylinders: Steel, Double Acting Piston Type with Hardened, Ground (Honed) and Chromed Rods

2.9. Ram speed: Multi speed ram is electrically controlled and allows for shift to lower speed before contacting sheet metal.

2.10. Ram Tilt Adjustment: Ram tilt shall be programmable through CNC controls and/or manual adjustment.

2.11. Ram level: The ram self-levels at the bottom of each stroke even if the hydraulic system is out of adjustment. Parallelism is maintained between bed and arm with

tolerance of +/- 0.0004" inch in low speed. The ram stops at the point of reference and self-levels on every stroke.

2.12. Ram Return: The ram shall have capability of auto return to upper travel limit.

Ram retraction shall have the capability of speed adjustment.

2.13. Bed referenced control: Monitors the actual distance between ram and bed, assuring repeatability independent of die wear or end plate deflection.

2.14. Front support arms: Quantity of two (2), 24" long, 4" wide front support arms with imbedded scales. Steel rule with 1/16" and 1mm increments shall be attached to each arm. T-slots with disappearing stops (help to hold the material from sliding backward).

2.15. Back-gauge system:

2.15.1. A slide and clamp, minimum 2 axis back-gauge.

2.15.2. CNC control operated back-gauge with digital readout (0.01" tolerance)

2.15.3. Back-gauge travel: 36" maximum

2.16. Stroke Control: Machine allows stroke control by rotation of hand wheels and/or CNC programmable control, which is read in thousandths of an inch and set the closed height. When the machine is 2 or 3 speed, the knob for setting the speed change point is located on right side for simple adjustments and/or CNC controlled.

2.17. Tonnage Control: adjustable tonnage control to go from lower tonnage to rated tonnage of machine, as well as overload protection. Tonnage control shall be CNC controlled and/or manually adjustable control.

2.18. Pendant Controller:

2.18.1. Pendant Controller is attached to a movable arm that extends from the machine.

2.18.2. Pendant Controller contains:

2.18.2.1. Power Front Operated Back-gauge

2.18.2.2. Programmable (GO TO) Function

2.18.2.3. Power Front Operated Back Gauge is motorized with a fine adjustment handwheel and/or CNC control.

2.18.2.4. The Ram Depth Control is motorized with a fine adjustment handwheel and/or CNC control.

2.18.3. The pendant controller shall have the following elements:

2.18.3.1. On/off switch (Pull to start, Push to stop, with Light on when in the "ON" position).

2.18.3.2. Run/Jog switch. Run position for normal operation. Jog position for loading of tooling, maintenance and set-up.

2.18.3.3. Up/Down buttons are used when in jog.

2.19. Foot control switch: The machine shall be equipped with a portable, electric three position foot control switch to cycle the machine. The foot control switch can be relocated from any position in front of the press brake or the control area.

2.20. Coolant: Oil cooler shall be provided with for the lubrication system.

2.21. Hydraulic system:

2.21.1. Provides minimum operating pressure 2500 psi

2.21.2. Automatic lubrication

2.22. Electric supply: 220V/440V, 3 Phase, 60 Hz.

2.23. Safety features:

2.23.1. All electrical components in power panels comply with NFPA 70/79 standards.

2.23.2. Machines are properly labeled with hazard points and comply with ANSI Z535.4

2.23.3. Machine safety guidelines comply with ANSI B11.15

2.23.4. Light curtain (Infrared)

2.23.5. Work light

2.23.6. Emergency stop button on pendant

2.24. Facility Connections: Electrical connections made to the facility electrical system shall conform to NFPA 70/79.

2.25. Enclosures: Components of the equipment shall be contained in an enclosure(s) of structural and sheet steel. Provisions shall be made for power cable entrance. The enclosure(s) shall be of drip-proof construction and of minimum size consistent with good design practices and ventilation of components. Electrical enclosures shall conform to NEMA 250, NFPA 79, NFPA 70 and NEMA 12.

2.26. Motors: All motors shall meet the requirements of NEMA MG 1

2.27. Caution – Warning Plates: Corrosion resistant "Caution" or "Warning" plates shall be securely attached to the equipment in prominent, visible locations. All safety precautions to be observed by the operator or maintenance personnel shall be permanently marked on the plates.

2.28. Identification Plate: A corrosion resistant identification plate shall be securely and conspicuously placed on the equipment. It shall contain, as a minimum, the following information:

2.28.1. Nomenclature

2.28.2. Contractor's Name

2.28.3. Manufacturer's Model Designation

2.28.4. Manufacturer's Serial Number

2.28.5. Electrical Utilities (Volts, H.P., Amps, Frequency, Phases, etc.)

2.28.6. Other Utilities (GPM, CFM, PSI, etc.), as applicable

2.28.7. Contract Number

2.28.8. Date of Manufacture

2.29. Operator/Repair/Maintenance Manuals: The equipment shall be furnished with three (3) hard copies and/or digital copies manufacturer's standard Operation, Maintenance, and Repair Manual(s).

2.29.1. A technical manual set is required to cover each specific make, model year, and serial numbered piece of equipment scheduled for delivery under the terms of the contract.

2.29.2. The manual shall provide instruction, illustrations, and other associated data for operations, maintenance, repair, overhaul, including a complete catalog of parts used in the assembly of the end item.

2.29.3. The manual provided shall contain complete instructions and information as set forth below for all equipment components, assemblies, subassemblies, attachments, and accessories purchased by the prime contractor from other sources and assembled in the end item.

2.29.4. Manuals shall be provided to the Technical Point of Contact at least 30 days prior to delivery.

2.29.5. The manuals shall include all mechanical and electrical schematics showing discrete components/block diagrams/wiring diagrams with inputs and outputs identified/system electrical interface documents and drawings for the specific model of all machine equipment/drives/controls supplied.

2.30. Drawings: Not later than 30 days after effective date of contract, Certified English language installation drawings shall be provided, in the manufacturer's standard format. An electronic copy shall also be provided in PDF format. The drawings shall contain at least, but shall not be limited to, the following information:

a. Overall and principal dimensions in sufficient detail to establish:

(1) limits of space in all directions required for installation, operation and servicing.

(2) Clearance for travel or rotation of any moving parts.

b. Interface mounting and mating information, such as dimensions of location for attaching hardware and physical footprint of equipment.

c. Interface pipe and cable attachments required for installation and co-functioning of the item to be installed with related items.

d. Information necessary for preparation of foundation plans, including mounting place details, hardware requirement lists, drilling plans and shock mounting buffers details.

- e. Weight of unit.
 - f. Total electrical requirements for machines (direct/indirect) to include Voltage, Amperage and cycles per second (Hz).
 - g. Location, type and dimensions of cable entrances, terminal boards and electrical connectors.
 - h. Interconnecting and cable detail.
3. **TECHNICAL MANUALS:** Three (3) complete sets of technical manuals covering operation, service and maintenance, and part-listing shall be provided.

4. **QUALITY ASSURANCE PROVISIONS** – Inspection and acceptance of the supplies, performed at destination by Technical Point of Contact (TPOC) and shall consist of type and kind; condition; operability, if readily determinable; preservation; packaging.

4.1 RESPONSIBILITY FOR INSPECTION: The contractor shall be responsible for the performance of all inspection requirements specified herein. The contractor may use his own facility or any other commercial facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The machine and all ancillary equipment shall completely conform to the requirements specified herein prior to the contractor requesting origin inspection from the Government. The Government reserves the right to perform any inspections set forth in the purchase description where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 INSPECTION / ACCEPTANCE TESTING (DESTINATION): An inspection and operational acceptance test shall be performed on the machine at destination to ensure conformance with this purchase description. The machine shall pass the examination, all tests, and the final inspection to be accepted. The supplies offered for delivery shall be examined for design, construction, components, electrical equipment and workmanship to determine conformance with the requirements of this specification. An operational test shall be conducted and witnessed by receiving activity following start up services. The machine shall be operated in a manner and for a time frame suitable to properly demonstrate and verify the successful operation of the equipment and all its component parts, related to the design, construction and performance. Included in this test, but not limited to, shall be modes of operation, all indicating devices, operator controls, hydraulic system, safety devices, and all elements of the control system.

4.3 RESPONSIBILITY FOR COMPLIANCE: All items must meet the requirements of section 2.

5. **ADDITIONAL REQUIREMENTS:**

5.1. Machine Hardware: Machines shall include any necessary lifting, leveling, and floor attachment hardware for machine installation.

5.2. Electrical Items: All electrical items shall meet the requirements of the current issue of NFPA 79 or equivalent (UL 508A).

5.3. Electric Power Disconnect: A lockable power feed disconnect device shall be provided and installed on the equipment to provide means to isolate power to machine.

5.4. CAUTION - WARNING PLATES: Corrosion resistant "Caution" or "Warning" plates shall be securely attached to the equipment in prominent, visible locations. All safety precautions to be observed by the operator or maintenance personnel shall be permanently marked on the plates.

5.5. Nameplate: A nameplate shall be permanently and securely attached to each machine the information provided on the plate shall be as listed below.

- a. Nomenclature
- b. Manufacturer's name
- c. Manufacturer's model designation
- d. Manufacturer's serial number
- e. Power input (volts, total amps, phase, frequency)
- f. Amp rating of largest motor
- g. Short circuit/over-current protection rating
- h. Contract Number or Order Number
- i. National Stock Number or Plant Equipment Code
- j. Date of manufacture

5.6. Operating and Maintenance Manual: Machines shall be provided with operating and Maintenance Manual (O&M), Electrical Manual, Mechanical Manual & Schematics three (3) each shall be provided with the machine and one (1) each searchable electronic copy (PDF) shall be E-mailed to TPOC.

5.7. Verification Test: After installation, the Contractor shall test run machine to ensure conformance with this purchase description.

6. Installation:

6.1. Installation/Foundation Drawings: Not later than sixty (60) calendar days after effective date of contract; installation requirements and drawings shall be provided to Norfolk Naval Shipyard TPOC.

6.2. Location: Machines shall be installed inside Bldg.234 1st floor NNSY (Norfolk Naval Shipyard, Portsmouth, VA 23709).

6.3. Existing Machine Removal: Norfolk Naval Shipyard will disconnect utilities and remove existing Press Break from installation location.

6.4. Mechanical Installation: The Contractor shall provide all labor, fluids, (except for coolant) piping, interconnecting hoses, and any other materials to move, locate, set

level, align, lubricate, and make ready to operate all equipment required by this contract. All machinery shall be installed to the original manufacturer's tolerances.

7.5 Available Utilities: 220/460 VAC, 3 phase, 60 Hz

7.6 Electrical Installation: The Contractor shall provide all labor, wiring, cables, disconnects, connectors, breakers and any other materials for complete installation of any necessary electrical equipment and electrically connect machine to building.

7.7 Start-up Services: Contractor shall provide start-up services for newly installed equipment.

7.8 SAFETY REQUIREMENTS: The Contractor shall develop and maintain a safety program in compliance with U.S. Army Corps of Engineers Safety and Health Requirements Manual EM 385-1-1, Public Law 91-596 Occupational Safety and Health Act, OSHA standards, 29 CFR 1910, 29 CFR 1915, and NAVSHIPYDNORINST P 5100.56, VOLUME II, CHAPTER 4. Contractor personnel shall work in a safe manner and comply with all applicable safety regulations. The Contractor shall be subject to safety inspections of its work sites by the Government. Contractor safety records shall be available upon request by the TPOC. The contractor shall be responsible for its Subcontractor's compliance with all of the aforementioned requirements.

7.8.1 MEDICAL TREATMENT: Only emergency medical care is available on Government facilities to Contractor employees who suffer on-the-job injury or disease. Care will be rendered under the conditions and at the rates in effect at the time of treatment. The Contractor shall reimburse the Naval Regional Medical Center Collection Agent promptly upon receipt of statement.

7.8.2 DANGER TAG/S AND LOCKS: Contractors and Non-NNSY Government Agencies (NNGAs) will work to NNSY's Lockout/Tags-Plus (LOTP) program in accordance with NAVSEA OSHECM Chapter 250 Rev. I, and OSHA 29 CFR 1915.89, "Hazardous Energy Control". Contractor/NNGA employees will affix appropriate lockout or Tags-Plus systems to energy isolating devices, and to otherwise disable machines, equipment, or systems (MES) to prevent unexpected energization, start up or release of hazardous energy in order to prevent injury to employees. For copies of OSHE 250, contact TPOC.

7.8.2.1 Before using appropriate Lockout or Tags-Plus systems, the contractor or NNGAs will provide a submittal confirming that all of their on-site personnel have been trained on the OSHECM Chapter 250 Lockout/Tags-Plus program, and 1915.89.

7.8.2.2 Contractors and NNGA employees will coordinate with NNSY designated Lockout/Tags-Plus Coordinators (LOTCs) and/or

Facility Custodians before applying any HEC lock or Tags-Plus systems to machinery, equipment, and/or systems. Hazardous Energy Control locks will be provided by the contractor or NNGAs for their employees' use. Hazardous Energy Control (HEC) tags will be provided by the cognizant NNSY LOTC if MES cannot be locked out. All HEC work under Tags-Plus controls will have an energy control procedure (ECP) directing hazardous energy controls for that work.

- 8 **Training:** The contractor shall provide training to familiarize Norfolk Naval Shipyard personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The training shall be performed at the customer facility between 7:20 am and 3:50 pm excluding any federal holidays.

8.1 Operator / Programmer On-Site Training: Training shall be provided at the receiving activity. Training shall include preparation of part programs, and preparation of equipment for safe operation. The trainees shall be provided with instruction for all aspects of machining operations. Trainees shall be provided with instruction for the care, use and operation of all attachments and accessories. Training shall be provided for three (3) operators/programmers. The training period shall not be greater than four (4) consecutive, eight (8) hour workdays for each machine. A training manual (which could be an additional operator's manual) shall be provided for each student.

8.2 Maintenance Personnel: Maintenance training shall include trouble-shooting and methods of correction if the equipment malfunctions, with particular emphasis on minimizing equipment down time. Training shall be provided for three (3) maintenance personnel. Training shall cover troubleshooting and maintain the machine. The training period shall not be less than one (1) eight (8) hour workday for each machine.

9 **FACILITIES ENTRY REQUIREMENTS:**

9.1 Initial Entry: Contractor provide a SECNAV 5512 and a CNRMA Installation Access Screening Request form that will be provided by the technical point of contact AT LEAST two (2) weeks prior to arrival at NNSY. These documents will be endorsed by the TPOC and forwarded to the NNSY Pass Office. Contractor shall submit documents no less than two weeks before arrival.

9.2 Vehicle Entry: The company name shall be displayed on each door of the Contractor's vehicle in a manner and size that is clearly visible. All vehicles shall display a valid state license plate that complies with the State Vehicle Code. Vehicles shall meet all other requirements of the State Vehicle Code, such as safety standards, and shall carry proof of insurance and state registration. Passes will be issued for vehicles to enter the shipyard. A separate pass is required for entering the Controlled Industrial Area (CIA).

9.3 Portable Electronic Device (PED): In accordance with COMSUBLANT/COMSUBPACINST 2075.1C, the use of Portable Electronic Devices (PEDs) while in or on Department of Navy Controlled Industrial Areas are prohibited. A portable electronic device (PED) is defined as any non-stationary electronic apparatus with singular or multiple capabilities of, but not limited to recording, storing, and/or transmitting data, voice, video or photo images (e.g., laptops, tablets, USB drives and wearable devices such as fitness bands and smart watches).

10 Critical time Constraints: The Installation and training shall be performed between the hours of 7:20 am and 3:50 pm excluding any federal holidays.

INSTALLATION RESPONSIBILITIES

<u>Government</u>	<u>Contractor</u>	
()	(XX)	a. Provide machine foundation in accordance with the manufacturer's design requirements.
()	(XX)	b. Furnish labor and material handling equipment for off-loading and placing items on foundation/designated area.
()	(XX)	c. Provide and install anchor bolts and nuts.
()	(XX)	d. Set and rough level the machine on its foundation.
()	(XX)	e. Level and align machine.
()	(XX)	f. Connect machine to the provided utilities hook-up.
()	(XX)	g. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests.
()	(XX)	h. Provide and charge all systems with fluids in accordance with manufacturer's instructions.
(XX)	()	i. Material for performance testing at Government facility.
()	(XX)	j. Material for performance testing at Contractor's facility.
(XX)	()	h. Provide utilities hook-up within 20 feet of the machine